Enhancing the Art Experience: Designing Immersive VR Paintings

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***Abstract*— In the realm of art museums and galleries, the engagement of tech-savvy young adults with traditional physical paintings poses a notable challenge. This challenge emanates from the perception held by these individuals, who regard physical artworks as static and incongruent with their preference for interactive digital media. To address this issue, the present study endeavors to mitigate this disparity by designing and developing "CanvasVerse," an immersive Virtual Reality application to bridge the disconnect between younger audiences and conventional paintings. The research primarily focusses on the investigation of CanvasVerse's usability and learnability within the demographic of young adults in Malaysia. Particularly, emphasis on understanding how interactivity and immersion, an integral characteristic of immersive VR experience, contribute positively to the appreciation of paintings. The empirical findings derived from System Usability Scale indicate that CanvasVerse effectively enhances emotional connections and engagement levels with physical paintings. Consequently, it provides a transformative encounter that also enhances the viewers' comprehension of artworks. These findings substantiate the potential of immersive VR technology to rekindle the appreciation of traditional paintings among younger demographics, thus marking a significant progression within the domain of art engagement and education. The study's contribution extends to the broader art community by introducing innovative avenues for engaging young audiences as well as deepening their understanding and connection with physical artworks.**

Keywords—Immersive VR Paintings, Traditional Paintings, System Usability Scale, Art Experience

I. Introduction

Engaging young adults to appreciate physical paintings is a significant challenge for museums and art galleries [1]. This is because this tech-savvy generation regards physical paintings as static, inaccessible, and irrelevant in the age of technology and social media. Unlike their predecessors, today's youth have grown up in a digital world, which has profoundly influenced their interactions with and perceptions of art. The traditional one-way viewing experience provided by physical paintings in galleries falls short of young audiences' expectations for participatory and interactive encounters. The exacerbation of this disparity is rooted in the evolving definitions of modern art among young individuals, often misaligned with traditional artistic criteria. Consequently, art galleries struggle with engaging this demographic with traditional physical artworks, prompting the pursuit of innovative solutions.

The challenges posed by physical paintings for younger generations are multifaceted. The passive and static nature of the viewing experience fails to resonate with tech-savvy young adults who increasingly prefer interactive digital content that encourages active engagement and participation. This absence of interaction leads to disconnection and diminished interest, impeding their art appreciation. Moreover, the divergence in societal landscapes complicates young people's ability to relate to the historical and cultural contexts depicted in conventional artworks. These historical narratives and symbolism often do not align with the contemporary sensibilities and trends of today's youth. Additionally, understanding physical paintings can prove daunting, particularly for those lacking a background in art history or formal art education. The complex nature of artistic techniques, symbolism, and contextual significance poses formidable barriers that hinder young adults from fully grasping the artistic merit and conveyed messages.

Researchers have explored a spectrum of technologies to enhance the appeal of physical paintings and re-engage young audiences. Collaborating closely with artists, they have orchestrated immersive and interactive encounters with artworks, harnessing technologies such as interactive displays, Augmented Reality (AR), haptic feedback, Virtual Reality (VR) exhibitions, and immersive VR. These technological interventions empower viewers to immerse themselves within the artworks, exploring diverse perspectives, engaging with virtual elements, and accessing supplementary information. Collectively, these features elevated levels of engagement, fostering dynamic and interactive experiences that amplify the comprehension and enjoyment of paintings. Among these technological innovations, immersive VR stands out as particularly promising, as it enables viewers to enter an immersive environment derived from the artworks themselves. Furthermore, immersive VR actively encourages participation in the artistic narrative, affording viewers the opportunity to cultivate profound emotional connections and heightened engagement through interactive elements, virtual exploration, and dynamic experiential encounters. Moreover, the educational potential of immersive VR is notable, as it provides interactive learning experiences for young audiences to delve deeper into art history, techniques, and the artist's intentions. Thus, making paintings more relatable and understandable to young adults.

Few studies have been conducted to investigate how immersive VR can transform the traditional art viewing experience. However, despite its enormous potential, the impact and acceptance of immersive VR technology on painting appreciation has remain relatively unexplored in existing literature. Therefore, the primary goal of the research is to design and develop "CanvasVerse," an immersive VR application that allows viewers to enter the canvas of Sulaiman Esa's paintings and become an integral part of the artistic narrative, with the goal of cultivating deeper emotional connections and increasing overall engagement through a transformative encounter with the paintings. This study aims to investigate the usability and learnability of CanvasVerse among young adults in Malaysia. This research work seeks to bridge the gap in the literature by investigating the acceptance of an immersive VR application on viewers' appreciation of physical paintings. The empirical evidence derived from the system usability scale evaluation assesses the effectiveness of application, shedding light on how interactivity and immersion positively influence painting appreciation. This study contributes to the field of VR technology in the art world, unlocking innovative avenues for engaging young audiences with physical paintings. The following sections of the paper are organized as follows: Section II reviews relevant literature on technologies that improve painting appreciation and art education; Section III provides comprehensive details on the methodology of designing and developing CanvasVerse; Section IV outlines the user evaluation methodology; and Section V discusses the findings. Finally, Section VI concludes the paper and outlining potential future research directions.

II. RELATED WORK

Painting has always been a powerful platform of expressing oneself. Recent technological advancements have opened new avenues to elevate how paintings are experienced and appreciated, particularly in academic settings such as art galleries and classrooms. These emerging technologies provide previously unimaginable new perspectives on how viewers can interact with and comprehend artworks. In this section, recent technologies that are being used to improve art appreciation and education are discussed, specifically interactive touchscreens, AR, haptic feedback, VR exhibitions, and immersive VR. These technologies allow for more immersive and interactive interactions with paintings.

Researchers have focused their efforts on exploring the use of interactive touchscreen technology to provide viewers with engaging and informative experiences. For example, through large multitouch tabletop, viewers can access background information about the painting through hand gesture [2]. Smaller touchscreen like tablet was also adopted to appreciate and engage with traditional Chinese painting [3]. Beyond touchscreens, AR technology was utilized to breathe new life into classic paintings by superimpose interactive digital content such as animations, audio, and contextual information on physical paintings. This approach transforms static paintings into a dynamic and informative encounter, bridging the gap between traditional art and modern audiences. For example, the use of mobile AR-guide system in art museum revealed that the AR technology significantly enhanced viewers' learning effectiveness, promoted a more immersive flow experience, and extended the time spent focusing on the paintings compared to audio and nonguided modes [4]. Besides, AR was also utilized to provide 3D virtual restorations of information necessary for understanding court paintings that supports historical context experiences of royal events and it was reported that AR provides visitors with a more immersive experience and gives more accurate information about royal events compared to the existing method [5]. There are also researchers investigating the use of haptic feedback technology as a means of adding a sense of touch and tactile sensations to the viewing experience. This method enhances the multisensory experience of art appreciation, making the interaction with the artwork more immersive and engaging. For instance, vibrations and ultrasound technology were utilized to generate haptic feedback that simulate the sensation of walking on paintings, creating a more engaging, immersive, and interactive experience for the viewer [6]. To offer viewers a more immersive journey into the world of art remotely and experience paintings in a more interactive manner, VR exhibitions was proposed. Through VR exhibition, visitors are transported into the artist's studio where they can view paintings. Interestingly, a study was conducted on the VR exhibition through desktop VR, head-mounted display (HMD) VR and physical paintings, and the findings indicate no significant difference in painting evaluation, indicating that participants perceived paintings, irrespective of VR or physical form, to be similar [7]. Thus, VR exhibition is suitable for viewers with mobility issues or those located in remote areas to experience renowned artworks from anywhere.

With the advancement of VR technology and drastic drop of HMD price in the recent years, researchers have started to adopt immersive VR to provide a transformative experience in the context of painting appreciation. Immersive VR was utilized to reinforce the relationship between art and history by putting the viewers in a meaningful context [8]. In another studies, students were provided with a more immersive and interactive experience to observe, explore the aesthetic spirit of art works, and ultimately promote art core literacy course [9]. Researchers also investigated the use of immersive VR to enhance art appreciation by making art more accessible and enjoyable to users [10]. The use of technologies presents remarkable opportunities to enhance the public's appreciation of paintings. However, the experience offered by immersive VR goes beyond traditional methods of viewing, fostering cultural understanding and emotional connections with artworks. Despite these immense possibilities, research on immersive VR technology on painting appreciation remains relatively unexplored. Specifically, investigation on the use of immersive VR technology to allow viewers to step inside the canvas and become an integral part of the artistic narrative to experience the painting. By addressing this research gap, full potential of immersive VR in enriching the world of art will be unlocked and bringing it closer to a wider audience.

III. DESIGN AND DEVELOPMENT OF CANVAVERSE

This section provides a detailed explanation of the development processes of CanvasVerse through Design Science Research (DSR), the selection of HMD, the intricate content creation workflows, and the strategic implementation of interactive scripting to deliver an immersive and engaging experience for viewers.

A. Design Science Research

Leveraging the DSR approach, CanvasVerse, an immersive VR application was conceived, designed, and developed. The aim is to address the contemporary challenge of revitalizing young adults' connection with physical artworks amidst evolving digital art consumption patterns [1]. The DSR methodology is a research approach dedicated to developing and evaluating innovative solutions to real-world problems and it serves as the study's guiding framework [11]. The DSR process started with the identification of prevalent challenges encountered by art galleries in their efforts to connect with young adult visitors. To better understand the issue's domain, an in-depth interview was conducted with Sulaiman Esa, a prominent artist from Malaysia. The outcomes of this interview unveiled a noticeable disconnection between young adults and traditional physical paintings displayed in galleries, resulting in a diminished appreciation for conventional artworks within this demographic. Having comprehended the problem, the study proceeds to formulate a succinct objective: the creation of an immersive VR application that immerses viewers within the canvas of Sulaiman Esa's artworks. This initiative enables them to actively engage in the artistic narrative. The core aim is to foster emotional bonds and enhance overall involvement with the paintings. This approach is designed to establish a sense of presence and immersion, thereby transforming the customary experience of viewing paintings into a dynamic and interactive encounter.

Following the objective-setting phase, a close collaboration ensued between the research team and the artist, resulting in the solicitation of invaluable insights and prerequisites. These inputs significantly shaped the conceptual design of CanvasVerse. The application was designed to provide viewers an array of captivating features, encompassing diverse perspectives for exploring paintings, interaction with virtual elements, and access to supplementary information regarding the paintings' historical context and symbolic connotations. The development phase involved the creation of immersive environments for each painting, intricate design of three-dimensional models, incorporation of non-player characters to enrich the storytelling, seamless integration of interactive components, and the crafting of user-friendly interfaces aimed at ensuring a seamless user experience. Upon completion of the development phase, CanvasVerse was demonstrated to the artist, Sulaiman Esa, who provided invaluable feedback to further refine the application. The iterative nature inherent in the DSR methodology enabled continuous enhancements, ensuring congruence between the final VR application and both the artist's artistic vision and the overarching goal of reigniting young audiences' engagement with physical paintings. Finally, a user evaluation was conducted with a targeted group of young adults to assess CanvasVerse's effectiveness and acceptance. The assessment's objective was to gauge the application's efficacy in enhancing the appreciation of physical paintings and evoking emotional connections among participants. This empirical results from the evaluation aims to elucidate how immersive VR technology, as exemplified by CanvasVerse, positively reshapes young audiences' perception and engagement with traditional paintings showcased within art galleries.

B. Selection of Head-Mounted Display

The choice of HMD bears significant implications for delivering an immersive yet captivating experience to viewers. Meta Quest 2 HMD was identified as the optimal choice to serve as the developmental and deployment platform for CanvasVerse. The selection is predicated on its cost-effectiveness and widespread accessibility. The wireless attributes of the Meta Quest 2 HMD, coupled with its support for six degrees of freedom tracking, amplify the naturalness and intuitiveness of viewer interactions within the VR environment [4]. By permitting viewers to explore the painting environment seamlessly and instinctively, the HMD augments engagement and overall satisfaction. Notably, the Meta Quest 2 HMD is complemented by the inclusion of Meta Touch controllers. These controllers, equipped with user-friendly buttons, triggers, and haptic feedback mechanisms, accentuate the intuitive and precision-driven tracking experience, thereby enriching the immersive encounter with the paintings.

C. Development of CanvaVerse

The developmental process of the CanvasVerse application, leading up to its deployment on the Meta Quest 2 HMD, is visually depicted in Fig. 1. The immersive realm of CanvasVerse springs to life through the creation of application assets, predominantly achieved using Maya and Blender 3D. Maya was chosen as the principal modelling tool due to its robust capabilities in crafting visually captivating three-dimensional models. Furthermore, Maya was utilized in creating a lifelike VR environment that authentically encapsulates the essence of the paintings themselves. In-depth research was conducted to uphold the authenticity of the virtual environments and accurately extend them in consonance with the artist's perspective. This pursuit of authenticity was augmented through insightful interviews with the artist himself, Sulaiman Esa. These interactions provided invaluable insights into the historical narratives, stories, and techniques woven into the paintings. The purpose is to ensure the faithful and respectful portrayal of the artworks within the immersive realm of VR.

To ensure a fluid and intuitive navigation experience, graphical user interfaces (GUI) were developed using illustrator and strategically positioned within the VR environment. Employing clarity and simplicity as guiding principles, the GUIs provide concise pathfinding guide, thereby facilitating seamless viewer navigation. The intricate task of generating materials and textures for the three-dimensional models, was executed using Blender 3D. This step involved deploying diverse strategies to replicate the paintings while safeguarding their authenticity. A prime illustration can be found in the reproduction of "View of Ayer Molek I", which has vibrant gouache hues and

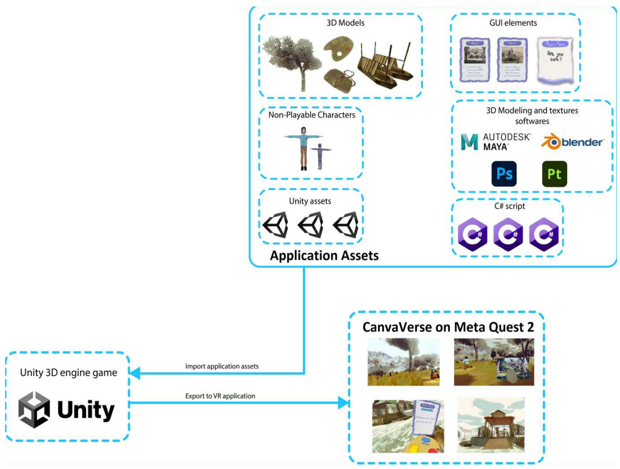

Fig. 1. Development processes for CanvasVerse

discernible brushstrokes, were directly derived from the artwork, thus, preserving the inherent texture and brushstroke intricacies. Conversely, a distinct approach was adopted for "Wet Market in JB City," a watercolor painting imbued with a sketch-like, unfinished appearance. This necessitated a hybrid approach, integrating manual texture creation with minimal extraction from the painting itself, thereby capturing its unique characteristics.

To infuse a sense of human presence and enhance viewer immersion, Maya was harnessed beyond the realm of static 3D assets to model non-player characters. These characters were thoughtfully animated and seamlessly woven into the immersive VR painting environment. This integration serves to further elevate the overall sense of immersion, effectively bridging the gap between the virtual and the physical paintings. The seamless integration of developed application assets and the VR environment into a cohesive experience was realized through the utilization of Unity3D for further development and coding. This choice was rooted in Unity3D's versatile profile as a game engine and its comprehensive toolset for crafting interactive and engaging VR landscapes [12]. Within Unity3D, the combination of 3D models, non-player characters, and the designed VR environment paved the way for constructing the immersive narrative of CanvasVerse. The first-person perspective was tailored to provide viewers with an embodiment of physical presence within the art space.

Complementary to visual elements, auditory dimensions played a vital role in heightening the overall ambiance of CanvasVerse. Rigorous selection of background music and sound effects was undertaken to align with the immersive encounter. These sound effects were strategically attuned to specific interactive components, effectively enhancing viewers' sense of engagement and interactivity. To enhance viewer engagement and interactivity, custom scripts were coded to handle interactive logic, empowering viewers to interact with the components within CanvasVerse to foster dynamic engagement. For example, in the Ayer Molek scene, viewers were required to find and interactively pick up pieces of paper, augmenting the immersion and agency within the artistic narrative, as shown in Fig. 2.

Upon the initiation of the VR CanvasVerse application, viewers will find themselves within the virtual confines of Sulaiman Esa's digitally recreated bedroom, an environment adorned with digital representations of his paintings. Fig. 3 depicts the digital replica of “Ayer Molek 1” and “Wet Market in JB City” of Sulaiman Esa. A key aspect of the immersive experience is the viewer's ability to navigate the virtual space with ease, facilitated by the teleportation function, an integral feature tailored to alleviate potential cybersickness, a common concern in VR environments. This design choice is aimed at ensuring a seamless and comfortable interaction.

For viewers seeking a deeper connection with the paintings, the CanvasVerse offers an enticing option of stepping into the paintings themselves. As an illustrative example, consider the painting "View of Ayer Molek I." A mere click by the viewer engenders a transformation, as they are instantaneously transported into the heart of the painting's virtual rendition as shown in Fig. 4. What ensues is an immersive journey, curated by the artist himself, wherein the painting evolves into a storytelling medium. Furthermore, the interaction extends beyond the canvas, as the viewer is granted an opportunity to engage in dialogue with the painter, Sulaiman Esa. This direct exchange, facilitated through the virtual environment, endows the viewer with insights into the historical and cultural context underpinning the artwork. Moreover, the Canvasverse extends beyond the realm of observation, inviting viewer to be an active participant. A noteworthy instance is embodied in the experience themed around "Wet Market in JB City." In this VR painting experience, the viewer is transported back in time to the 1960s, specifically to the bustling wet market adjacent to the Segget river in Johor Baru. In this setting, the viewer is placed within a small boat, imbued with a palpable ambiance of commotion, authentic sounds of sellers and crowds permeate the air. This integration of multisensory components which include visual, auditory, and narrative, crafted an all-encompassing immersion, allowing viewer to glean an experiential understanding of the historical wet market and the everyday lives of Johorians of that era as depicted in Fig. 5.

Fig. 2. Viewer Locating and Picking Up Pieces of Paper

Fig. 3a. Ayer Molek 1 Fig. 3b. Wet Market in JB City

Fig. 3. Digital Masterpieces: Virtual Replicas of Sulaiman Esa's Paintings

IV. User Evaluation

In order to elicit valuable insights into the usability and learnability of the novel VR application named "CanvasVerse," a preliminary assessment was carried out employing a quantitative methodology cantered around the System Usability Scale (SUS), as developed by Brooke [13].

Fig. 4. Stepping into the VR Canvas: "View of Ayer Molek I"

Fig. 5. Journey to the Past: Immersive Glimpse of 1960s Wet Market

The SUS, a well-established tool, encompasses a comprehensive questionnaire comprising of ten items. Respondents rate these items on a 5-point Likert scale, ranging from 1 (indicating strong disagreement) to 5 (indicating strong agreement). This instrument has been widely adopted across various domains, including VR applications [14, 15], as it offers a swift and user-friendly means to evaluate usability with a modest sample size. SUS scores are normalized on a scale ranging from 0 (representing the lowest usability) to 100 (indicating the highest usability), thereby furnishing a quantitative gauge of user-perceived usability [16]. Significantly, extant research attests to the credibility and consistency of the SUS as a tool for appraising usability [17], affirming its robustness and adaptability for usability practitioners. The primary inquiry addressed by this investigation revolves around the reception of VR as a transformative platform for artwork appreciation among young adults. To seek answers to the query, a convenience sampling approach was employed to recruit a cohort of 17 participants aged between 18 and 24 years. The participants include 5 females and 12 males with varying levels of experience with VR technology. The participants were not remunerated for their involvement in the study. It is pertinent to note that the selection of Malaysian as the study locale was not driven by an intent to scrutinize local contextual factors. Rather, Malaysian was chosen to contribute to the broad spectrum of data representation, engendering diversity, and not to elucidate country-specific dynamics or assert claims regarding unique Malaysian attributes.

The evaluation was conducted in conjunction of UNLOCK 2023, an esteemed Graduation Show hosted by Faculty of Creative Multimedia, Multimedia University, Cyberjaya, Malaysia. Each participant was given a briefing of the evaluation's objectives, the functionalities of CanvasVerse, and the testing procedures. Subsequent to this, participants provided their consent via Google Forms before embarking on the evaluation. Proficiency in handling the Meta Quest 2 HMD was ensured through a preparatory tutorial provided to each participant. An immersive encounter with CanvasVerse within a fully operational demonstration environment was facilitated, with participants allocated approximately 25 minutes for this engagement. Guided tasks were assigned to participants to navigate and experience the CanvaVerse application. Following the evaluation interaction, participants were directed to a Google Form encompassing a 10-item SUS questionnaire. Following the evaluation phase, a question-and-answer session was held to obtain additional qualitative insights from the participants.

V. DISCUSSION

The primary goal of this evaluation was to examine the acceptance of CanvasVerse, a platform that harnesses immersive VR technology to cultivate profound emotional connections and heightened engagement with paintings. Incorporating gender as a variable within the user evaluation was prompted by the recognition that individuals' perspectives and reactions to immersive VR encounters may exhibit variations depending on their gender identities. The individual SUS scores obtained from each participant, which range from 52.5 to 100, are depicted in Fig. 6. The evaluation results reveal a mean SUS score of 85.44, with a median value of 87.50. These metrics collectively denote a commendable level of usability pertaining to the CanvasVerse VR application. The analysis of SUS scores revealed notable gender-based differences in usability perceptions. The female participants exhibited a higher mean SUS score of 93.00, indicating that the design and content choices of CanvaVerse may have catered more effectively as perceived by female participants. In contrast, the male participants, while still expressing a positive acceptance with a mean SUS score of 82.30, rated slightly lower in comparison. Although the findings suggest that the emotional and immersive aspects of CanvasVerse to appreciate art, may have resonated more profoundly with the female participants, leading to a higher degree of acceptance among this demographic, the male participants also expressed positive evaluations, implying that CanvasVerse possesses broad appeal.

Individual responses in the form of mean and median from the participants for each of the SUS questions are provided in Table 1, drawing meaningful insights that contribute to a holistic understanding of the acceptance of CanvaVerse among young adults. The positive response with a mean score of 4.41 and a median of 5.00 to Q1, underscores the participants' strong inclination to immerse themselves in CanvasVerse repeatedly, demonstrating its appeal among the target audience. The findings from Q2, with a mean score of 1.65 and a median of 1.00, suggest that the participants did not perceive CanvasVerse as unduly complex. The low mean score is crucial in the context of usability, as an uncomplicated user experience is pivotal for engagement. Q3 reveals a high positive perception among the participants, with a mean score of 4.47 and a median of 5.00. This suggests that the participants found CanvasVerse to be user-friendly and intuitive, which resonates with the objective of the research to provide an accessible and engaging VR art experience. The responses to Q4 and Q8, both reflect positively on CanvasVerse. With mean scores of 2.12 and 1.47, respectively, the participants did not strongly

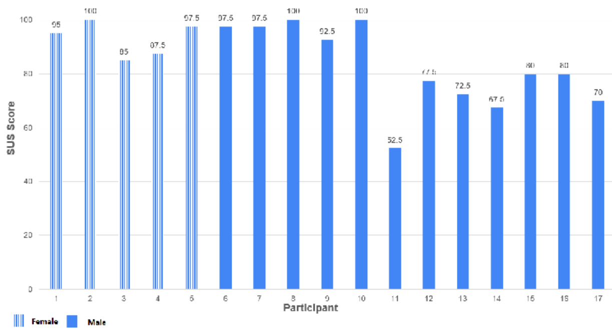

Fig. 6. SUS score per participant for the CanvaVerse

feel the need for assistance or perceive the application as cumbersome. These responses further affirm the user-friendly nature of CanvasVerse. For Q5 and Q6, the mean score of 4.71 and 1.53 indicates that the participants perceived the functions within CanvasVerse as well-integrated and did not find significant inconsistencies. These findings validate the application's effectiveness in delivering a cohesive and engaging VR art experience. The responses to Q7 shed light on the perceived learnability of CanvasVerse, with a mean score of 4.24. The participants generally believed that most individuals would swiftly learn to use the application, indicating a short learning curve and underscoring the application's potential for wide acceptance. Q9 garners a mean score of 4.71, indicating that the participants felt very confident using CanvasVerse. Q10 elicits positive responses, with a mean score of 1.59. This imply that the participants did not require arduous learning prior to using CanvaVerse to appreciate VR art experience.

TABLE I. EVALUATION RESULT: INDIVIDUAL SUS QUESTIONS

SUS Question		Mean	Median
Q1	I think that I would like to use the CanvaVerse frequently.	4.41	5.00
Q2	I found this CanvaVerse unnecessarily complex.	1.65	1.00
Q3	I thought this CanvaVerse was easy to use.	4.47	5.00
Q4	I think that I would need assistance to be able to use this CanvaVerse.	2.12	2.00
Q5	I found the various functions in this CanvaVerse were well integrated.	4.71	5.00
Q6	I thought there was too much inconsistency in this CanvaVerse.	1.53	1.00
Q7	I would imagine that most people would learn to use this CanvaVerse very quickly.	4.24	4.00
Q8	I found this CanvaVerse very cumbersome to use.	1.47	1.00
Q9	I felt very confident using this CanvaVerse.	4.71	5.00
Q10	I needed to learn a lot of things before I could get going with the CanvaVerse.	1.59	2.00

VI. CONCLUSION

In the domain of art engagement, where art galleries grapple persistently with the challenge of connecting younger adult demographic and traditional physical artworks, notably within the tech-savvy segment. This study introduced CanvasVerse, an immersive VR application, engineered to enhance the level of appreciation among young adults for the realm of paintings. Through SUS evaluation results, the research found compelling evidence that CanvasVerse effectively fosters emotional connections and engagement with physical paintings among its users, with an overall SUS score of 85.44 and positive responses to specific usability aspects. The evaluation results harmonize seamlessly with the objective of the study, which is to delve into the manner in which interactivity and immersion of immersive VR exert a favorable influence on the appreciation of paintings. The implications of these findings are important, serving as compelling demonstrations of immersive VR technology's capacity to reignite interest in conventional artworks, making them more accessible and captivating to the younger adults. By incorporating immersive VR into their exhibitions, art galleries stand to gain significantly, opening new channels for engaging their audiences, deepening their understanding of artistic works, and expand beyond the confines of physical limitations. Nonetheless, it is important to acknowledge the study's limitations, which include its specific cultural and geographical context, and to emphasize the need for further research into long-term effects and potential adoption barriers. Future research should focus on these aspects in order to fully understand the potential of immersive VR in enriching the realm of art and fostering greater inclusivity for future generations. In short, CanvasVerse represents a noteworthy advancement in the field of art engagement and education. It effectively showcases the transformative potential inherent in immersive VR technology, revitalising interest in traditional painting appreciation, and adeptly bridging the gap between the realms of art and technology, particularly among younger generations.

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